1.4 **[0,6점]** τ

$$\delta Q = \alpha \pi R_E^2 \tau \quad (10)$$

$$\tau = \frac{M}{\alpha \pi R_E^2 \mu_{air}} \frac{\gamma R \Delta T}{\gamma - 1} = 30.3 \cdot 10^3 \text{ s}. \quad (11)$$

제2 부. 단열 대기 (6,8점)

2.1 **[1,2점]** ,
(5)

$$\rho = \frac{\mu_{air} p}{RT}. \quad (12)$$

$$p T^{\frac{\gamma}{1-\gamma}} = const. \quad (13)$$

(4) (12) (13)

$$\frac{dT}{dh} = -\frac{(\gamma-1)\mu_{air} g}{\gamma R} = -\beta = const. \quad (14)$$

(14)

$$T(h) = T_0 - \frac{(\gamma-1)\mu_{air} g}{\gamma R} h = T_0 - \beta h \quad (15)$$

$H = 1500\ m$

$$T(H) = 278 \text{ K}. \quad (16)$$

2.2 **[0,4점]**
(13)

$$p(h) = p_0 \left(\frac{T_0}{T(h)}\right)^{\frac{\gamma}{1-\gamma}} = p_0 \left(\frac{T_0}{T_0 - \beta h}\right)^{\frac{\gamma}{1-\gamma}} \quad (17)$$

$H = 1500\ m$

$$p(H) = 84.6 \cdot 10^3\ Pa. \quad (18)$$

2.3 **[0,8점]** ,
(15)

$$T(h) = T_0 - \beta h = const. \quad (19)$$

$$\Delta H_{atm} = \frac{\gamma R \Delta T_{dn}}{(\gamma-1)\mu_{air} g} = 2{,}05 \cdot 10^3 \text{ m}. \quad (20)$$

2.4 **[0,6점]**

$$p(T) = p_1 + \frac{p_2 - p_1}{T_2 - T_1}(T - T_1). \quad (21)$$

,

.

(18) (21)

$$T_{boil} = 368 \text{ K}. \quad (22)$$

2.5 **[0,8점]** ,
0 ℃ . ,

$$T_{melt} = 273 \text{ K}. \quad (23)$$

(15)

$$h_0 = \frac{\gamma R (T_0 - T_{melt})}{(\gamma-1)\mu_{air} g} = 2.05 \cdot 10^3 \text{ m}. \quad (24)$$

2.6 **[0,4점]**

. (24)

$$H_0 = \frac{\gamma R (T - T_{melt})}{(\gamma-1)\mu_{air} g} = 3.78 \cdot 10^3 \text{ m}. \quad (25)$$